

Package ‘FeatureExtraction’

June 26, 2026

Type Package

Title Generating Features for a Cohort

Version 3.14.0

Date 2026-06-25

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Description An R interface for generating features for a cohort using data in the Common Data Model. Features can be constructed using default or custom made feature definitions. Furthermore it's possible to aggregate features and get the summary statistics.

Depends R (>= 3.2.2),
DatabaseConnector (>= 3.0.0),
Andromeda (>= 1.0.0)

Imports methods,
dplyr,
rJava,
jsonlite,
SqlRender (>= 1.18.0),
ParallelLogger (>= 2.0.2),
cli,
pillar,
readr,
rlang,
RSQLite,
DBI,
checkmate,
vroom

Suggests testthat,
knitr,
rmarkdown,
Eunomia (>= 2.0.0),
withr,
curl,
http

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VignetteBuilder knitr

URL <https://github.com/OHDSI/FeatureExtraction>

BugReports <https://github.com/OHDSI/FeatureExtraction/issues>

NeedsCompilation no

RoxygenNote 7.3.3

Encoding UTF-8

Language en-US

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```
.createLooCovariateSettings
```

Get covariate settings

Description

Get covariate settings

Usage

```
.createLooCovariateSettings(useLengthOfObs = TRUE)
```

Arguments

`useLengthOfObs` if length of observations should be used

Value

Returns an object of type `covariateSettings`, containing settings for the covariates.

```
.getDbLooCovariateData
```

Get covariate information from the database

Description

Get covariate information from the database

Usage

```
.getDbLooCovariateData(  
  connection,  
  tempEmulationSchema = NULL,  
  cdmDatabaseSchema,  
  cohortTable = "#cohort_person",  
  cohortIds = c(-1),  
  cdmVersion = "5",  
  rowIdField = "subject_id",  
  covariateSettings,  
  aggregated = FALSE,  
  minCharacterizationMean = 0,  
  minCharacterizationCount = 0,  
  ...  
)
```

Arguments

connection	A connection to the server containing the schema as created using the connect function in the DatabaseConnector package. Either the connection or connectionDetails argument should be specified.
tempEmulationSchema	Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.
cdmDatabaseSchema	The name of the database schema that contains the OMOP CDM instance. Requires read permissions to this database. On SQL Server, this should specify both the database and the schema, so for example 'cdm_instance.dbo'.
cohortTable	Name of the (temp) table holding the cohort for which we want to construct covariates
cohortIds	For which cohort ID(s) should covariates be constructed? If set to -1, covariates will be constructed for all cohorts in the specified cohort table.
cdmVersion	Define the OMOP CDM version used: currently supported is "5".
rowIdField	The name of the field in the cohort table that is to be used as the row_id field in the output table. This can be especially usefull if there is more than one period per person.
covariateSettings	Either an object of type covariateSettings as created using one of the create-Covariate functions, or a list of such objects.
aggregated	Should aggregate statistics be computed instead of covariates per cohort entry?
minCharacterizationMean	The minimum mean value for binary characterization output. Values below this will be cut off from output. This will help reduce the file size of the characterization output, but will remove information on covariates that have very low values. The default is 0.
minCharacterizationCount	The minimum count value for binary characterization output. Values below this will be cut off from output. This will help reduce the file size of the characterization output, but will remove information on covariates that occur in very few cohort entries. The default is 0.
...	Additional arguments, not used.

Value

Returns an object of type covariateData, containing information on the covariates.

aggregateCovariates	<i>Aggregate covariate data</i>
---------------------	---------------------------------

Description

Aggregate covariate data

Usage

```
aggregateCovariates(covariateData)
```

Arguments

`covariateData` An object of type `covariateData` as generated using `getDbCovariateData`.

Value

An object of class `covariateData`.

Examples

```
covariateData <- FeatureExtraction::createEmptyCovariateData(  
  cohortIds = 1,  
  aggregated = FALSE,  
  temporal = FALSE  
)  
aggregatedCovariateData <- aggregateCovariates(covariateData)
```

`computeStandardizedDifference`

Compute standardized difference of mean for all covariates.

Description

Computes the standardized difference for all covariates between two cohorts. The standardized difference is defined as the difference between the mean divided by the overall standard deviation.

Usage

```
computeStandardizedDifference(  
  covariateData1,  
  covariateData2,  
  cohortId1 = NULL,  
  cohortId2 = NULL  
)
```

Arguments

`covariateData1` The covariate data of the first cohort. Needs to be in aggregated format.

`covariateData2` The covariate data of the second cohort. Needs to be in aggregated format.

`cohortId1` If provided, `covariateData1` will be restricted to this cohort. If not provided, `covariateData1` is assumed to contain data on only 1 cohort.

`cohortId2` If provided, `covariateData2` will be restricted to this cohort. If not provided, `covariateData2` is assumed to contain data on only 1 cohort.

Value

A data frame with means and standard deviations per cohort as well as the standardized difference of mean.

Examples

```
binaryCovDataFile <- system.file("testdata/binaryCovariateData.zip",
  package = "FeatureExtraction"
)
covariateData1 <- loadCovariateData(binaryCovDataFile)
covariateData2 <- loadCovariateData(binaryCovDataFile)
covDataDiff <- computeStandardizedDifference(
  covariateData1,
  covariateData2,
  cohortId1 = 1,
  cohortId2 = 2
)
```

convertPrespecSettingsToDetailedSettings

Convert prespecified covariate settings into detailed covariate settings

Description

Convert prespecified covariate settings into detailed covariate settings

Usage

```
convertPrespecSettingsToDetailedSettings(covariateSettings)
```

Arguments

covariateSettings

An object of type covariateSettings as created for example by the [createCovariateSettings](#) function.

Details

For advanced users only.

Value

An object of type covariateSettings, to be used in other functions.

Examples

```
covSettings <- createDefaultCovariateSettings()
detailedSettings <- convertPrespecSettingsToDetailedSettings(covariateSettings = covSettings)
```

CovariateData-class *Covariate Data*

Description

CovariateData is an S4 class that inherits from [Andromeda](#). It contains information on covariates, which can be either captured on a per-person basis, or aggregated across the cohort(s).

By default covariates refer to a specific time period, with for example different covariate IDs for whether a diagnosis code was observed in the year before and month before index date. However, a CovariateData can also be temporal, meaning that next to a covariate ID there is also a time ID, which identifies the (user specified) time window the covariate was captured.

A CovariateData object is typically created using [getDbCovariateData](#), can only be saved using [saveCovariateData](#), and loaded using [loadCovariateData](#).

Usage

```
## S4 method for signature 'CovariateData'
show(object)

## S4 method for signature 'CovariateData'
summary(object)
```

Arguments

object An object of class ‘CovariateData’.

See Also

[isCovariateData](#), [isAggregatedCovariateData](#), [isTemporalCovariateData](#)

createAnalysisDetails *Create detailed covariate settings*

Description

Create detailed covariate settings

Usage

```
createAnalysisDetails(
  analysisId,
  sqlFileName,
  parameters,
  includedCovariateConceptIds = c(),
  addDescendantsToInclude = FALSE,
  excludedCovariateConceptIds = c(),
  addDescendantsToExclude = FALSE,
  includedCovariateIds = c()
)
```

Arguments

analysisId	An integer between 0 and 999 that uniquely identifies this analysis.
sqlFileName	The name of the parameterized SQL file embedded in the featureExtraction package.
parameters	The list of parameter values used to render the template SQL.
includedCovariateConceptIds	A list of concept IDs that should be used to construct covariates.
addDescendantsToInclude	Should descendant concept IDs be added to the list of concepts to include?
excludedCovariateConceptIds	A list of concept IDs that should NOT be used to construct covariates.
addDescendantsToExclude	Should descendant concept IDs be added to the list of concepts to exclude?
includedCovariateIds	A list of covariate IDs that should be restricted to.

Details

creates an object specifying in detail how covariates should be constructed from data in the CDM model. Warning: this function is for advanced users only.

Value

An object of type analysisDetail, to be used in [createDetailedCovariateSettings](#) or [createDetailedTemporalC](#)

Examples

```
analysisDetails <- createAnalysisDetails(
  analysisId = 1,
  sqlFileName = "DemographicsGender.sql",
  parameters = list(
    analysisId = 1,
    analysisName = "Gender",
    domainId = "Demographics"
  ),
  includedCovariateConceptIds = c(),
  addDescendantsToInclude = FALSE,
  excludedCovariateConceptIds = c(),
  addDescendantsToExclude = FALSE,
  includedCovariateIds = c()
)
```

createCohortAttrCovariateSettings

Create cohort attribute covariate settings

Description

Create cohort attribute covariate settings

Usage

```
createCohortAttrCovariateSettings(
  analysisId = -1,
  attrDatabaseSchema,
  attrDefinitionTable = "attribute_definition",
  cohortAttrTable = "cohort_attribute",
  includeAttrIds = c(),
  isBinary = FALSE,
  missingMeansZero = FALSE
)
```

Arguments

analysisId A unique identifier for this analysis.

attrDatabaseSchema The database schema where the attribute definition and cohort attribute table can be found.

attrDefinitionTable The name of the attribute definition table.

cohortAttrTable The name of the cohort attribute table.

includeAttrIds (optional) A list of attribute definition IDs to restrict to.

isBinary Needed for aggregation: Are these binary variables? Binary variables should only have the values 0 or 1.

missingMeansZero Needed for aggregation: For continuous values, should missing values be interpreted as 0?

Details

Creates an object specifying where the cohort attributes can be found to construct covariates. The attributes should be defined in a table with the same structure as the `attribute_definition` table in the Common Data Model. It should at least have these columns:

attribute_definition_id A unique identifier of type integer.

attribute_name A short description of the attribute.

The cohort attributes themselves should be stored in a table with the same format as the `cohort_attribute` table in the Common Data Model. It should at least have these columns:

cohort_definition_id A key to link to the cohort table.

subject_id A key to link to the cohort table.

cohort_start_date A key to link to the cohort table.

attribute_definition_id An foreign key linking to the attribute definition table.

value_as_number A real number.

Value

An object of type `covariateSettings`, to be used in other functions.

Examples

```

covariateSettings <- createCohortAttrCovariateSettings(
  analysisId = 1,
  attrDatabaseSchema = "main",
  attrDefinitionTable = "attribute_definition",
  cohortAttrTable = "cohort_attribute",
  includeAttrIds = c(1),
  isBinary = FALSE,
  missingMeansZero = FALSE
)

```

```
createCohortBasedCovariateSettings
```

Create settings for covariates based on other cohorts

Description

Create settings for covariates based on other cohorts

Usage

```

createCohortBasedCovariateSettings(
  analysisId,
  covariateCohortDatabaseSchema = NULL,
  covariateCohortTable = NULL,
  covariateCohorts,
  valueType = "binary",
  startDay = -365,
  endDay = 0,
  includedCovariateIds = c(),
  warnOnAnalysisIdOverlap = TRUE
)

```

Arguments

<code>analysisId</code>	A unique identifier for this analysis.
<code>covariateCohortDatabaseSchema</code>	The database schema where the cohorts used to define the covariates can be found. If set to NULL, the database schema will be guessed, for example using the same one as for the main cohorts.
<code>covariateCohortTable</code>	The table where the cohorts used to define the covariates can be found. If set to NULL, the table will be guessed, for example using the same one as for the main cohorts.
<code>covariateCohorts</code>	A data frame with at least two columns: 'cohortId' and 'cohortName'. The cohort ID should correspond to the cohort_definition_id of the cohort to use for creating a covariate.

valueType	Either 'binary' or 'count'. When valueType = 'count', the covariate value will be the number of times the cohort was observed in the window.
startDay	What is the start day (relative to the index date) of the covariate window?
endDay	What is the end day (relative to the index date) of the covariate window?
includedCovariateIds	A list of covariate IDs that should be restricted to.
warnOnAnalysisIdOverlap	Warn if the provided 'analysisId' overlaps with any predefined analysis as available in the 'createCovariateSettings()' function.

Details

Creates an object specifying covariates to be constructed based on the presence of other cohorts.

Value

An object of type covariateSettings, to be used in other functions.

```
createCohortBasedTemporalCovariateSettings
```

Create settings for temporal covariates based on other cohorts

Description

Create settings for temporal covariates based on other cohorts

Usage

```
createCohortBasedTemporalCovariateSettings(
  analysisId,
  covariateCohortDatabaseSchema = NULL,
  covariateCohortTable = NULL,
  covariateCohorts,
  valueType = "binary",
  temporalStartDays = -365:-1,
  temporalEndDays = -365:-1,
  includedCovariateIds = c(),
  warnOnAnalysisIdOverlap = TRUE
)
```

Arguments

analysisId	A unique identifier for this analysis.
covariateCohortDatabaseSchema	The database schema where the cohorts used to define the covariates can be found. If set to NULL, the database schema will be guessed, for example using the same one as for the main cohorts.
covariateCohortTable	The table where the cohorts used to define the covariates can be found. If set to NULL, the table will be guessed, for example using the same one as for the main cohorts.

covariateCohorts	A data frame with at least two columns: 'cohortId' and 'cohortName'. The cohort ID should correspond to the cohort_definition_id of the cohort to use for creating a covariate.
valueType	Either 'binary' or 'count'. When valueType = 'count', the covariate value will be the number of times the cohort was observed in the window.
temporalStartDays	A list of integers representing the start of a time period, relative to the index date. 0 indicates the index date, -1 indicates the day before the index date, etc. The start day is included in the time period.
temporalEndDays	A list of integers representing the end of a time period, relative to the index date. 0 indicates the index date, -1 indicates the day before the index date, etc. The end day is included in the time period.
includedCovariateIds	A list of covariate IDs that should be restricted to.
warnOnAnalysisIdOverlap	Warn if the provided 'analysisId' overlaps with any predefined analysis as available in the 'createTemporalCovariateSettings()' function.

Details

Creates an object specifying temporal covariates to be constructed based on the presence of other cohorts.

Value

An object of type covariateSettings, to be used in other functions.

createCovariateSettings

Create covariate settings

Description

Create covariate settings

Usage

```
createCovariateSettings(
  useDemographicsGender = FALSE,
  useDemographicsAge = FALSE,
  useDemographicsAgeGroup = FALSE,
  useDemographicsRace = FALSE,
  useDemographicsEthnicity = FALSE,
  useDemographicsIndexYear = FALSE,
  useDemographicsIndexMonth = FALSE,
  useDemographicsPriorObservationTime = FALSE,
  useDemographicsPostObservationTime = FALSE,
  useDemographicsTimeInCohort = FALSE,
  useDemographicsIndexYearMonth = FALSE,
```

```
useCareSiteId = FALSE,
useConditionOccurrenceAnyTimePrior = FALSE,
useConditionOccurrenceLongTerm = FALSE,
useConditionOccurrenceMediumTerm = FALSE,
useConditionOccurrenceShortTerm = FALSE,
useConditionOccurrencePrimaryInpatientAnyTimePrior = FALSE,
useConditionOccurrencePrimaryInpatientLongTerm = FALSE,
useConditionOccurrencePrimaryInpatientMediumTerm = FALSE,
useConditionOccurrencePrimaryInpatientShortTerm = FALSE,
useConditionEraAnyTimePrior = FALSE,
useConditionEraLongTerm = FALSE,
useConditionEraMediumTerm = FALSE,
useConditionEraShortTerm = FALSE,
useConditionEraOverlapping = FALSE,
useConditionEraStartLongTerm = FALSE,
useConditionEraStartMediumTerm = FALSE,
useConditionEraStartShortTerm = FALSE,
useConditionGroupEraAnyTimePrior = FALSE,
useConditionGroupEraLongTerm = FALSE,
useConditionGroupEraMediumTerm = FALSE,
useConditionGroupEraShortTerm = FALSE,
useConditionGroupEraOverlapping = FALSE,
useConditionGroupEraStartLongTerm = FALSE,
useConditionGroupEraStartMediumTerm = FALSE,
useConditionGroupEraStartShortTerm = FALSE,
useDrugExposureAnyTimePrior = FALSE,
useDrugExposureLongTerm = FALSE,
useDrugExposureMediumTerm = FALSE,
useDrugExposureShortTerm = FALSE,
useDrugEraAnyTimePrior = FALSE,
useDrugEraLongTerm = FALSE,
useDrugEraMediumTerm = FALSE,
useDrugEraShortTerm = FALSE,
useDrugEraOverlapping = FALSE,
useDrugEraStartLongTerm = FALSE,
useDrugEraStartMediumTerm = FALSE,
useDrugEraStartShortTerm = FALSE,
useDrugGroupEraAnyTimePrior = FALSE,
useDrugGroupEraLongTerm = FALSE,
useDrugGroupEraMediumTerm = FALSE,
useDrugGroupEraShortTerm = FALSE,
useDrugGroupEraOverlapping = FALSE,
useDrugGroupEraStartLongTerm = FALSE,
useDrugGroupEraStartMediumTerm = FALSE,
useDrugGroupEraStartShortTerm = FALSE,
useProcedureOccurrenceAnyTimePrior = FALSE,
useProcedureOccurrenceLongTerm = FALSE,
useProcedureOccurrenceMediumTerm = FALSE,
useProcedureOccurrenceShortTerm = FALSE,
useDeviceExposureAnyTimePrior = FALSE,
useDeviceExposureLongTerm = FALSE,
useDeviceExposureMediumTerm = FALSE,
```

```

useDeviceExposureShortTerm = FALSE,
useMeasurementAnyTimePrior = FALSE,
useMeasurementLongTerm = FALSE,
useMeasurementMediumTerm = FALSE,
useMeasurementShortTerm = FALSE,
useMeasurementValueAnyTimePrior = FALSE,
useMeasurementValueLongTerm = FALSE,
useMeasurementValueMediumTerm = FALSE,
useMeasurementValueShortTerm = FALSE,
useMeasurementRangeGroupAnyTimePrior = FALSE,
useMeasurementRangeGroupLongTerm = FALSE,
useMeasurementRangeGroupMediumTerm = FALSE,
useMeasurementRangeGroupShortTerm = FALSE,
useMeasurementValueAsConceptAnyTimePrior = FALSE,
useMeasurementValueAsConceptLongTerm = FALSE,
useMeasurementValueAsConceptMediumTerm = FALSE,
useMeasurementValueAsConceptShortTerm = FALSE,
useObservationAnyTimePrior = FALSE,
useObservationLongTerm = FALSE,
useObservationMediumTerm = FALSE,
useObservationShortTerm = FALSE,
useObservationValueAsConceptAnyTimePrior = FALSE,
useObservationValueAsConceptLongTerm = FALSE,
useObservationValueAsConceptMediumTerm = FALSE,
useObservationValueAsConceptShortTerm = FALSE,
useObservationValueAnyTimePrior = FALSE,
useObservationValueLongTerm = FALSE,
useObservationValueMediumTerm = FALSE,
useObservationValueShortTerm = FALSE,
useCharlsonIndex = FALSE,
useDcsi = FALSE,
useChads2 = FALSE,
useChads2Vasc = FALSE,
useHfrs = FALSE,
useDistinctConditionCountLongTerm = FALSE,
useDistinctConditionCountMediumTerm = FALSE,
useDistinctConditionCountShortTerm = FALSE,
useDistinctIngredientCountLongTerm = FALSE,
useDistinctIngredientCountMediumTerm = FALSE,
useDistinctIngredientCountShortTerm = FALSE,
useDistinctProcedureCountLongTerm = FALSE,
useDistinctProcedureCountMediumTerm = FALSE,
useDistinctProcedureCountShortTerm = FALSE,
useDistinctMeasurementCountLongTerm = FALSE,
useDistinctMeasurementCountMediumTerm = FALSE,
useDistinctMeasurementCountShortTerm = FALSE,
useDistinctObservationCountLongTerm = FALSE,
useDistinctObservationCountMediumTerm = FALSE,
useDistinctObservationCountShortTerm = FALSE,
useVisitCountLongTerm = FALSE,
useVisitCountMediumTerm = FALSE,
useVisitCountShortTerm = FALSE,

```

```

    useVisitConceptCountLongTerm = FALSE,
    useVisitConceptCountMediumTerm = FALSE,
    useVisitConceptCountShortTerm = FALSE,
    longTermStartDays = -365,
    mediumTermStartDays = -180,
    shortTermStartDays = -30,
    endDays = 0,
    includedCovariateConceptIds = c(),
    addDescendantsToInclude = FALSE,
    excludedCovariateConceptIds = c(),
    addDescendantsToExclude = FALSE,
    includedCovariateIds = c()
  )

```

Arguments

`useDemographicsGender`
Gender of the subject. (analysis ID 1)

`useDemographicsAge`
Age of the subject on the index date (in years). (analysis ID 2)

`useDemographicsAgeGroup`
Age of the subject on the index date (in 5 year age groups) (analysis ID 3)

`useDemographicsRace`
Race of the subject. (analysis ID 4)

`useDemographicsEthnicity`
Ethnicity of the subject. (analysis ID 5)

`useDemographicsIndexYear`
Year of the index date. (analysis ID 6)

`useDemographicsIndexMonth`
Month of the index date. (analysis ID 7)

`useDemographicsPriorObservationTime`
Number of continuous days of observation time preceding the index date. (analysis ID 8)

`useDemographicsPostObservationTime`
Number of continuous days of observation time following the index date. (analysis ID 9)

`useDemographicsTimeInCohort`
Number of days of observation time during cohort period. (analysis ID 10)

`useDemographicsIndexYearMonth`
Both calendar year and month of the index date in a single variable. (analysis ID 11)

`useCareSiteId` Care site associated with the cohort start, pulled from the `visit_detail`, `visit_occurrence`, or `person` table, in that order. (analysis ID 12)

`useConditionOccurrenceAnyTimePrior`
One covariate per condition in the `condition_occurrence` table starting any time prior to index. (analysis ID 101)

`useConditionOccurrenceLongTerm`
One covariate per condition in the `condition_occurrence` table starting in the long term window. (analysis ID 102)

`useConditionOccurrenceMediumTerm`
 One covariate per condition in the `condition_occurrence` table starting in the medium term window. (analysis ID 103)

`useConditionOccurrenceShortTerm`
 One covariate per condition in the `condition_occurrence` table starting in the short term window. (analysis ID 104)

`useConditionOccurrencePrimaryInpatientAnyTimePrior`
 One covariate per condition observed as a primary diagnosis in an inpatient setting in the `condition_occurrence` table starting any time prior to index. (analysis ID 105)

`useConditionOccurrencePrimaryInpatientLongTerm`
 One covariate per condition observed as a primary diagnosis in an inpatient setting in the `condition_occurrence` table starting in the long term window. (analysis ID 106)

`useConditionOccurrencePrimaryInpatientMediumTerm`
 One covariate per condition observed as a primary diagnosis in an inpatient setting in the `condition_occurrence` table starting in the medium term window. (analysis ID 107)

`useConditionOccurrencePrimaryInpatientShortTerm`
 One covariate per condition observed as a primary diagnosis in an inpatient setting in the `condition_occurrence` table starting in the short term window. (analysis ID 108)

`useConditionEraAnyTimePrior`
 One covariate per condition in the `condition_era` table overlapping with any time prior to index. (analysis ID 201)

`useConditionEraLongTerm`
 One covariate per condition in the `condition_era` table overlapping with any part of the long term window. (analysis ID 202)

`useConditionEraMediumTerm`
 One covariate per condition in the `condition_era` table overlapping with any part of the medium term window. (analysis ID 203)

`useConditionEraShortTerm`
 One covariate per condition in the `condition_era` table overlapping with any part of the short term window. (analysis ID 204)

`useConditionEraOverlapping`
 One covariate per condition in the `condition_era` table overlapping with the end of the risk window. (analysis ID 205)

`useConditionEraStartLongTerm`
 One covariate per condition in the `condition_era` table starting in the long term window. (analysis ID 206)

`useConditionEraStartMediumTerm`
 One covariate per condition in the `condition_era` table starting in the medium term window. (analysis ID 207)

`useConditionEraStartShortTerm`
 One covariate per condition in the `condition_era` table starting in the short term window. (analysis ID 208)

`useConditionGroupEraAnyTimePrior`
 One covariate per condition era rolled up to groups in the `condition_era` table overlapping with any time prior to index. (analysis ID 209)

useConditionGroupEraLongTerm	One covariate per condition era rolled up to groups in the condition_era table overlapping with any part of the long term window. (analysis ID 210)
useConditionGroupEraMediumTerm	One covariate per condition era rolled up to groups in the condition_era table overlapping with any part of the medium term window. (analysis ID 211)
useConditionGroupEraShortTerm	One covariate per condition era rolled up to groups in the condition_era table overlapping with any part of the short term window. (analysis ID 212)
useConditionGroupEraOverlapping	One covariate per condition era rolled up to groups in the condition_era table overlapping with the end of the risk window. (analysis ID 213)
useConditionGroupEraStartLongTerm	One covariate per condition era rolled up to groups in the condition_era table starting in the long term window. (analysis ID 214)
useConditionGroupEraStartMediumTerm	One covariate per condition era rolled up to groups in the condition_era table starting in the medium term window. (analysis ID 215)
useConditionGroupEraStartShortTerm	One covariate per condition era rolled up to groups in the condition_era table starting in the short term window. (analysis ID 216)
useDrugExposureAnyTimePrior	One covariate per drug in the drug_exposure table starting any time prior to index. (analysis ID 301)
useDrugExposureLongTerm	One covariate per drug in the drug_exposure table starting in the long term window. (analysis ID 302)
useDrugExposureMediumTerm	One covariate per drug in the drug_exposure table starting in the medium term window. (analysis ID 303)
useDrugExposureShortTerm	One covariate per drug in the drug_exposure table starting in the short term window. (analysis ID 304)
useDrugEraAnyTimePrior	One covariate per drug in the drug_era table overlapping with any time prior to index. (analysis ID 401)
useDrugEraLongTerm	One covariate per drug in the drug_era table overlapping with any part of the long term window. (analysis ID 402)
useDrugEraMediumTerm	One covariate per drug in the drug_era table overlapping with any part of the medium term window. (analysis ID 403)
useDrugEraShortTerm	One covariate per drug in the drug_era table overlapping with any part of the short term window. (analysis ID 404)
useDrugEraOverlapping	One covariate per drug in the drug_era table overlapping with the end of the risk window. (analysis ID 405)
useDrugEraStartLongTerm	One covariate per drug in the drug_era table starting in the long term window. (analysis ID 406)

useDrugEraStartMediumTerm
One covariate per drug in the drug_era table starting in the medium term window. (analysis ID 407)

useDrugEraStartShortTerm
One covariate per drug in the drug_era table starting in the short term window. (analysis ID 408)

useDrugGroupEraAnyTimePrior
One covariate per drug rolled up to ATC groups in the drug_era table overlapping with any time prior to index. (analysis ID 409)

useDrugGroupEraLongTerm
One covariate per drug rolled up to ATC groups in the drug_era table overlapping with any part of the long term window. (analysis ID 410)

useDrugGroupEraMediumTerm
One covariate per drug rolled up to ATC groups in the drug_era table overlapping with any part of the medium term window. (analysis ID 411)

useDrugGroupEraShortTerm
One covariate per drug rolled up to ATC groups in the drug_era table overlapping with any part of the short term window. (analysis ID 412)

useDrugGroupEraOverlapping
One covariate per drug rolled up to ATC groups in the drug_era table overlapping with the end of the risk window. (analysis ID 413)

useDrugGroupEraStartLongTerm
One covariate per drug rolled up to ATC groups in the drug_era table starting in the long term window. (analysis ID 414)

useDrugGroupEraStartMediumTerm
One covariate per drug rolled up to ATC groups in the drug_era table starting in the medium term window. (analysis ID 415)

useDrugGroupEraStartShortTerm
One covariate per drug rolled up to ATC groups in the drug_era table starting in the short term window. (analysis ID 416)

useProcedureOccurrenceAnyTimePrior
One covariate per procedure in the procedure_occurrence table any time prior to index. (analysis ID 501)

useProcedureOccurrenceLongTerm
One covariate per procedure in the procedure_occurrence table in the long term window. (analysis ID 502)

useProcedureOccurrenceMediumTerm
One covariate per procedure in the procedure_occurrence table in the medium term window. (analysis ID 503)

useProcedureOccurrenceShortTerm
One covariate per procedure in the procedure_occurrence table in the short term window. (analysis ID 504)

useDeviceExposureAnyTimePrior
One covariate per device in the device exposure table starting any time prior to index. (analysis ID 601)

useDeviceExposureLongTerm
One covariate per device in the device exposure table starting in the long term window. (analysis ID 602)

useDeviceExposureMediumTerm
One covariate per device in the device exposure table starting in the medium term window. (analysis ID 603)

useDeviceExposureShortTerm	One covariate per device in the device exposure table starting in the short term window. (analysis ID 604)
useMeasurementAnyTimePrior	One covariate per measurement in the measurement table any time prior to index. (analysis ID 701)
useMeasurementLongTerm	One covariate per measurement in the measurement table in the long term window. (analysis ID 702)
useMeasurementMediumTerm	One covariate per measurement in the measurement table in the medium term window. (analysis ID 703)
useMeasurementShortTerm	One covariate per measurement in the measurement table in the short term window. (analysis ID 704)
useMeasurementValueAnyTimePrior	One covariate containing the value per measurement-unit combination any time prior to index. (analysis ID 705)
useMeasurementValueLongTerm	One covariate containing the value per measurement-unit combination in the long term window. (analysis ID 706)
useMeasurementValueMediumTerm	One covariate containing the value per measurement-unit combination in the medium term window. (analysis ID 707)
useMeasurementValueShortTerm	One covariate containing the value per measurement-unit combination in the short term window. (analysis ID 708)
useMeasurementRangeGroupAnyTimePrior	Covariates indicating whether measurements are below, within, or above normal range any time prior to index. (analysis ID 709)
useMeasurementRangeGroupLongTerm	Covariates indicating whether measurements are below, within, or above normal range in the long term window. (analysis ID 710)
useMeasurementRangeGroupMediumTerm	Covariates indicating whether measurements are below, within, or above normal range in the medium term window. (analysis ID 711)
useMeasurementRangeGroupShortTerm	Covariates indicating whether measurements are below, within, or above normal range in the short term window. (analysis ID 712)
useMeasurementValueAsConceptAnyTimePrior	One covariate per measurement-value concept combination any time prior to index. (analysis ID 713)
useMeasurementValueAsConceptLongTerm	One covariate per measurement-value concept combination in the long term window. (analysis ID 714)
useMeasurementValueAsConceptMediumTerm	One covariate per measurement-value concept combination in the medium term window. (analysis ID 715)
useMeasurementValueAsConceptShortTerm	One covariate per measurement-value concept combination in the short term window. (analysis ID 716)

useObservationAnyTimePrior	One covariate per observation in the observation table any time prior to index. (analysis ID 801)
useObservationLongTerm	One covariate per observation in the observation table in the long term window. (analysis ID 802)
useObservationMediumTerm	One covariate per observation in the observation table in the medium term window. (analysis ID 803)
useObservationShortTerm	One covariate per observation in the observation table in the short term window. (analysis ID 804)
useObservationValueAsConceptAnyTimePrior	One covariate per observation-value concept combination any time prior to index. (analysis ID 805)
useObservationValueAsConceptLongTerm	One covariate per observation-value concept combination in the long term window. (analysis ID 806)
useObservationValueAsConceptMediumTerm	One covariate per observation-value concept combination in the medium term window. (analysis ID 807)
useObservationValueAsConceptShortTerm	One covariate per observation-value concept combination in the short term window. (analysis ID 808)
useObservationValueAnyTimePrior	One covariate containing the value per observation-unit combination any time prior to index. (analysis ID 809)
useObservationValueLongTerm	One covariate containing the value per observation-unit combination in the long term window. (analysis ID 810)
useObservationValueMediumTerm	One covariate containing the value per observation-unit combination in the medium term window. (analysis ID 811)
useObservationValueShortTerm	One covariate containing the value per observation-unit combination in the short term window. (analysis ID 812)
useCharlsonIndex	The Charlson comorbidity index (Romano adaptation) using all conditions prior to the window end. (analysis ID 901)
useDcsi	The Diabetes Comorbidity Severity Index (DCSI) using all conditions prior to the window end. (analysis ID 902)
useChads2	The CHADS2 score using all conditions prior to the window end. (analysis ID 903)
useChads2Vasc	The CHADS2VAsC score using all conditions prior to the window end. (analysis ID 904)
useHfrs	The Hospital Frailty Risk Score score using all conditions prior to the window end. (analysis ID 926)
useDistinctConditionCountLongTerm	The number of distinct condition concepts observed in the long term window. (analysis ID 905)

useDistinctConditionCountMediumTerm
The number of distinct condition concepts observed in the medium term window. (analysis ID 906)

useDistinctConditionCountShortTerm
The number of distinct condition concepts observed in the short term window. (analysis ID 907)

useDistinctIngredientCountLongTerm
The number of distinct ingredients observed in the long term window. (analysis ID 908)

useDistinctIngredientCountMediumTerm
The number of distinct ingredients observed in the medium term window. (analysis ID 909)

useDistinctIngredientCountShortTerm
The number of distinct ingredients observed in the short term window. (analysis ID 910)

useDistinctProcedureCountLongTerm
The number of distinct procedures observed in the long term window. (analysis ID 911)

useDistinctProcedureCountMediumTerm
The number of distinct procedures observed in the medium term window. (analysis ID 912)

useDistinctProcedureCountShortTerm
The number of distinct procedures observed in the short term window. (analysis ID 913)

useDistinctMeasurementCountLongTerm
The number of distinct measurements observed in the long term window. (analysis ID 914)

useDistinctMeasurementCountMediumTerm
The number of distinct measurements observed in the medium term window. (analysis ID 915)

useDistinctMeasurementCountShortTerm
The number of distinct measurements observed in the short term window. (analysis ID 916)

useDistinctObservationCountLongTerm
The number of distinct observations observed in the long term window. (analysis ID 917)

useDistinctObservationCountMediumTerm
The number of distinct observations observed in the medium term window. (analysis ID 918)

useDistinctObservationCountShortTerm
The number of distinct observations observed in the short term window. (analysis ID 919)

useVisitCountLongTerm
The number of visits observed in the long term window. (analysis ID 920)

useVisitCountMediumTerm
The number of visits observed in the medium term window. (analysis ID 921)

useVisitCountShortTerm
The number of visits observed in the short term window. (analysis ID 922)

useVisitConceptCountLongTerm
The number of visits observed in the long term window, stratified by visit concept ID. (analysis ID 923)

useVisitConceptCountMediumTerm	The number of visits observed in the medium term window, stratified by visit concept ID. (analysis ID 924)
useVisitConceptCountShortTerm	The number of visits observed in the short term window, stratified by visit concept ID. (analysis ID 925)
longTermStartDays	What is the start day (relative to the index date) of the long-term window?
mediumTermStartDays	What is the start day (relative to the index date) of the medium-term window?
shortTermStartDays	What is the start day (relative to the index date) of the short-term window?
endDays	What is the end day (relative to the index date) of the window?
includedCovariateConceptIds	A list of concept IDs that should be used to construct covariates.
addDescendantsToInclude	Should descendant concept IDs be added to the list of concepts to include?
excludedCovariateConceptIds	A list of concept IDs that should NOT be used to construct covariates.
addDescendantsToExclude	Should descendant concept IDs be added to the list of concepts to exclude?
includedCovariateIds	A list of covariate IDs that should be restricted to.

Details

creates an object specifying how covariates should be constructed from data in the CDM model.

Value

An object of type covariateSettings, to be used in other functions.

Examples

```
settings <- createCovariateSettings(
  useDemographicsGender = TRUE,
  useDemographicsAge = FALSE,
  useDemographicsAgeGroup = TRUE,
  useDemographicsRace = TRUE,
  useDemographicsEthnicity = TRUE,
  useDemographicsIndexYear = TRUE,
  useDemographicsIndexMonth = TRUE,
  useDemographicsPriorObservationTime = FALSE,
  useDemographicsPostObservationTime = FALSE,
  useDemographicsTimeInCohort = FALSE,
  useDemographicsIndexYearMonth = FALSE,
  useCareSiteId = FALSE,
  useConditionOccurrenceAnyTimePrior = FALSE,
  useConditionOccurrenceLongTerm = FALSE,
  useConditionOccurrenceMediumTerm = FALSE,
  useConditionOccurrenceShortTerm = FALSE,
  useConditionOccurrencePrimaryInpatientAnyTimePrior = FALSE,
  useConditionOccurrencePrimaryInpatientLongTerm = FALSE,
```

```
useConditionOccurrencePrimaryInpatientMediumTerm = FALSE,
useConditionOccurrencePrimaryInpatientShortTerm = FALSE,
useConditionEraAnyTimePrior = FALSE,
useConditionEraLongTerm = FALSE,
useConditionEraMediumTerm = FALSE,
useConditionEraShortTerm = FALSE,
useConditionEraOverlapping = FALSE,
useConditionEraStartLongTerm = FALSE,
useConditionEraStartMediumTerm = FALSE,
useConditionEraStartShortTerm = FALSE,
useConditionGroupEraAnyTimePrior = FALSE,
useConditionGroupEraLongTerm = TRUE,
useConditionGroupEraMediumTerm = FALSE,
useConditionGroupEraShortTerm = TRUE,
useConditionGroupEraOverlapping = FALSE,
useConditionGroupEraStartLongTerm = FALSE,
useConditionGroupEraStartMediumTerm = FALSE,
useConditionGroupEraStartShortTerm = FALSE,
useDrugExposureAnyTimePrior = FALSE,
useDrugExposureLongTerm = FALSE,
useDrugExposureMediumTerm = FALSE,
useDrugExposureShortTerm = FALSE,
useDrugEraAnyTimePrior = FALSE,
useDrugEraLongTerm = FALSE,
useDrugEraMediumTerm = FALSE,
useDrugEraShortTerm = FALSE,
useDrugEraOverlapping = FALSE,
useDrugEraStartLongTerm = FALSE,
useDrugEraStartMediumTerm = FALSE,
useDrugEraStartShortTerm = FALSE,
useDrugGroupEraAnyTimePrior = FALSE,
useDrugGroupEraLongTerm = TRUE,
useDrugGroupEraMediumTerm = FALSE,
useDrugGroupEraShortTerm = TRUE,
useDrugGroupEraOverlapping = TRUE,
useDrugGroupEraStartLongTerm = FALSE,
useDrugGroupEraStartMediumTerm = FALSE,
useDrugGroupEraStartShortTerm = FALSE,
useProcedureOccurrenceAnyTimePrior = FALSE,
useProcedureOccurrenceLongTerm = TRUE,
useProcedureOccurrenceMediumTerm = FALSE,
useProcedureOccurrenceShortTerm = TRUE,
useDeviceExposureAnyTimePrior = FALSE,
useDeviceExposureLongTerm = TRUE,
useDeviceExposureMediumTerm = FALSE,
useDeviceExposureShortTerm = TRUE,
useMeasurementAnyTimePrior = FALSE,
useMeasurementLongTerm = TRUE,
useMeasurementMediumTerm = FALSE,
useMeasurementShortTerm = TRUE,
useMeasurementValueAnyTimePrior = FALSE,
useMeasurementValueLongTerm = FALSE,
useMeasurementValueMediumTerm = FALSE,
useMeasurementValueShortTerm = FALSE,
useMeasurementRangeGroupAnyTimePrior = FALSE,
useMeasurementRangeGroupLongTerm = TRUE,
useMeasurementRangeGroupMediumTerm = FALSE,
```

```

useMeasurementRangeGroupShortTerm = TRUE,
useMeasurementValueAsConceptAnyTimePrior = FALSE,
useMeasurementValueAsConceptLongTerm = TRUE,
useMeasurementValueAsConceptMediumTerm = FALSE,
useMeasurementValueAsConceptShortTerm = TRUE,
useObservationAnyTimePrior = FALSE,
useObservationLongTerm = TRUE,
useObservationMediumTerm = FALSE,
useObservationShortTerm = TRUE,
useObservationValueAsConceptAnyTimePrior = FALSE,
useObservationValueAsConceptLongTerm = TRUE,
useObservationValueAsConceptMediumTerm = FALSE,
useObservationValueAsConceptShortTerm = TRUE,
useObservationValueAnyTimePrior = FALSE,
useObservationValueLongTerm = FALSE,
useObservationValueMediumTerm = FALSE,
useObservationValueShortTerm = FALSE,
useCharlsonIndex = TRUE,
useDcsi = TRUE,
useChads2 = TRUE,
useChads2Vasc = TRUE,
useHfrs = FALSE,
useDistinctConditionCountLongTerm = FALSE,
useDistinctConditionCountMediumTerm = FALSE,
useDistinctConditionCountShortTerm = FALSE,
useDistinctIngredientCountLongTerm = FALSE,
useDistinctIngredientCountMediumTerm = FALSE,
useDistinctIngredientCountShortTerm = FALSE,
useDistinctProcedureCountLongTerm = FALSE,
useDistinctProcedureCountMediumTerm = FALSE,
useDistinctProcedureCountShortTerm = FALSE,
useDistinctMeasurementCountLongTerm = FALSE,
useDistinctMeasurementCountMediumTerm = FALSE,
useDistinctMeasurementCountShortTerm = FALSE,
useDistinctObservationCountLongTerm = FALSE,
useDistinctObservationCountMediumTerm = FALSE,
useDistinctObservationCountShortTerm = FALSE,
useVisitCountLongTerm = FALSE,
useVisitCountMediumTerm = FALSE,
useVisitCountShortTerm = FALSE,
useVisitConceptCountLongTerm = FALSE,
useVisitConceptCountMediumTerm = FALSE,
useVisitConceptCountShortTerm = FALSE,
longTermStartDays = -365,
mediumTermStartDays = -180,
shortTermStartDays = -30,
endDays = 0,
includedCovariateConceptIds = c(),
addDescendantsToInclude = FALSE,
excludedCovariateConceptIds = c(),
addDescendantsToExclude = FALSE,
includedCovariateIds = c()
)

```

`createDefaultCovariateSettings`*Create default covariate settings*

Description

Create default covariate settings

Usage

```
createDefaultCovariateSettings(  
  includedCovariateConceptIds = c(),  
  addDescendantsToInclude = FALSE,  
  excludedCovariateConceptIds = c(),  
  addDescendantsToExclude = FALSE,  
  includedCovariateIds = c()  
)
```

Arguments

`includedCovariateConceptIds`
A list of concept IDs that should be used to construct covariates.

`addDescendantsToInclude`
Should descendant concept IDs be added to the list of concepts to include?

`excludedCovariateConceptIds`
A list of concept IDs that should NOT be used to construct covariates.

`addDescendantsToExclude`
Should descendant concept IDs be added to the list of concepts to exclude?

`includedCovariateIds`
A list of covariate IDs that should be restricted to.

Value

An object of type `covariateSettings`, to be used in other functions.

Examples

```
covSettings <- createDefaultCovariateSettings(  
  includedCovariateConceptIds = c(1),  
  addDescendantsToInclude = FALSE,  
  excludedCovariateConceptIds = c(2),  
  addDescendantsToExclude = FALSE,  
  includedCovariateIds = c(1)  
)
```

```
createDefaultTemporalCovariateSettings
```

Create default covariate settings

Description

Create default covariate settings

Usage

```
createDefaultTemporalCovariateSettings(  
  includedCovariateConceptIds = c(),  
  addDescendantsToInclude = FALSE,  
  excludedCovariateConceptIds = c(),  
  addDescendantsToExclude = FALSE,  
  includedCovariateIds = c()  
)
```

Arguments

includedCovariateConceptIds
A list of concept IDs that should be used to construct covariates.

addDescendantsToInclude
Should descendant concept IDs be added to the list of concepts to include?

excludedCovariateConceptIds
A list of concept IDs that should NOT be used to construct covariates.

addDescendantsToExclude
Should descendant concept IDs be added to the list of concepts to exclude?

includedCovariateIds
A list of covariate IDs that should be restricted to.

Value

An object of type `covariateSettings`, to be used in other functions.

Examples

```
covSettings <- createDefaultTemporalCovariateSettings(  
  includedCovariateConceptIds = c(1),  
  addDescendantsToInclude = FALSE,  
  excludedCovariateConceptIds = c(2),  
  addDescendantsToExclude = FALSE,  
  includedCovariateIds = c(1)  
)
```

```
createDetailedCovariateSettings
```

Create detailed covariate settings

Description

Create detailed covariate settings

Usage

```
createDetailedCovariateSettings(analyses = list())
```

Arguments

analyses A list of analysisDetail objects as created using [createAnalysisDetails](#).

Details

creates an object specifying in detail how covariates should be constructed from data in the CDM model. Warning: this function is for advanced users only.

Value

An object of type covariateSettings, to be used in other functions.

Examples

```
analysisDetails <- createAnalysisDetails(
  analysisId = 1,
  sqlFileName = "DemographicsGender.sql",
  parameters = list(
    analysisId = 1,
    analysisName = "Gender",
    domainId = "Demographics"
  ),
  includedCovariateConceptIds = c(),
  addDescendantsToInclude = FALSE,
  excludedCovariateConceptIds = c(),
  addDescendantsToExclude = FALSE,
  includedCovariateIds = c()
)
covSettings <- createDetailedCovariateSettings(analyses = analysisDetails)
```

```
createDetailedTemporalCovariateSettings
```

Create detailed temporal covariate settings

Description

Create detailed temporal covariate settings

Usage

```
createDetailedTemporalCovariateSettings(  
  analyses = list(),  
  temporalStartDays = -365:-1,  
  temporalEndDays = -365:-1  
)
```

Arguments

analyses A list of analysis detail objects as created using [createAnalysisDetails](#).

temporalStartDays A list of integers representing the start of a time period, relative to the index date. 0 indicates the index date, -1 indicates the day before the index date, etc. The start day is included in the time period.

temporalEndDays A list of integers representing the end of a time period, relative to the index date. 0 indicates the index date, -1 indicates the day before the index date, etc. The end day is included in the time period.

Details

creates an object specifying in detail how temporal covariates should be constructed from data in the CDM model. Warning: this function is for advanced users only.

Value

An object of type covariateSettings, to be used in other functions.

Examples

```
analysisDetails <- createAnalysisDetails(  
  analysisId = 1,  
  sqlFileName = "DemographicsGender.sql",  
  parameters = list(  
    analysisId = 1,  
    analysisName = "Gender",  
    domainId = "Demographics"  
  ),  
  includedCovariateConceptIds = c(),  
  addDescendantsToInclude = FALSE,  
  excludedCovariateConceptIds = c(),  
  addDescendantsToExclude = FALSE,  
  includedCovariateIds = c())
```

```

)
covSettings <- createDetailedTemporalCovariateSettings(
  analyses = analysisDetails,
  temporalStartDays = -365:-1,
  temporalEndDays = -365:-1
)

```

createEmptyCovariateData

Creates an empty covariate data object

Description

Creates an empty covariate data object

Usage

```
createEmptyCovariateData(cohortIds, aggregated, temporal)
```

Arguments

cohortIds	For which cohort IDs should the covariate data be created?
aggregated	if the data should be aggregated
temporal	if the data is temporary

Value

an empty object of class CovariateData

Examples

```

covariateData <- FeatureExtraction::createEmptyCovariateData(
  cohortIds = 1,
  aggregated = FALSE,
  temporal = FALSE
)

```

createTable1

Create a table 1

Description

Creates a formatted table of cohort characteristics, to be included in publications or reports. Allows for creating a table describing a single cohort, or a table comparing two cohorts.

Usage

```
createTable1(
  covariateData1,
  covariateData2 = NULL,
  cohortId1 = NULL,
  cohortId2 = NULL,
  specifications = getDefaultTable1Specifications(),
  output = "two columns",
  showCounts = FALSE,
  showPercent = TRUE,
  percentDigits = 1,
  valueDigits = 1,
  stdDiffDigits = 2
)
```

Arguments

covariateData1	The covariate data of the cohort to be included in the table.
covariateData2	The covariate data of the cohort to also be included, when comparing two cohorts.
cohortId1	If provided, covariateData1 will be restricted to this cohort. If not provided, covariateData1 is assumed to contain data on only 1 cohort.
cohortId2	If provided, covariateData2 will be restricted to this cohort. If not provided, covariateData2 is assumed to contain data on only 1 cohort.
specifications	Specifications of which covariates to display, and how.
output	The output format for the table. Options are output = "two columns", output = "one column", or output = "list".
showCounts	Show the number of cohort entries having the binary covariate?
showPercent	Show the percentage of cohort entries having the binary covariate?
percentDigits	Number of digits to be used for percentages.
valueDigits	Number of digits to be used for the values of continuous variables.
stdDiffDigits	Number of digits to be used for the standardized differences.

Value

A data frame, or, when output = "list" a list of two data frames.

Examples

```
eunomiaConnectionDetails <- Eunomia::getEunomiaConnectionDetails()
covSettings <- createDefaultCovariateSettings()
Eunomia::createCohorts(
  connectionDetails = eunomiaConnectionDetails,
  cdmDatabaseSchema = "main",
  cohortDatabaseSchema = "main",
  cohortTable = "cohort"
)
covData1 <- getDbCovariateData(
  connectionDetails = eunomiaConnectionDetails,
  tempEmulationSchema = NULL,
```

```

    cdmDatabaseSchema = "main",
    cdmVersion = "5",
    cohortTable = "cohort",
    cohortDatabaseSchema = "main",
    cohortTableIsTemp = FALSE,
    cohortId = 1,
    rowIdField = "subject_id",
    covariateSettings = covSettings,
    aggregated = TRUE
  )
  covData2 <- getDbCovariateData(
    connectionDetails = eunomiaConnectionDetails,
    tempEmulationSchema = NULL,
    cdmDatabaseSchema = "main",
    cdmVersion = "5",
    cohortTable = "cohort",
    cohortDatabaseSchema = "main",
    cohortTableIsTemp = FALSE,
    cohortId = 2,
    rowIdField = "subject_id",
    covariateSettings = covSettings,
    aggregated = TRUE
  )
  table1 <- createTable1(
    covariateData1 = covData1,
    covariateData2 = covData2,
    cohortId1 = 1,
    cohortId2 = 2,
    specifications = getDefaultTable1Specifications(),
    output = "one column",
    showCounts = FALSE,
    showPercent = TRUE,
    percentDigits = 1,
    valueDigits = 1,
    stdDiffDigits = 2
  )

```

createTable1CovariateSettings

Create covariate settings for a table 1

Description

Creates a covariate settings object for generating only those covariates that will be included in a table 1. This function works by filtering the covariateSettings object for the covariates in the specifications object.

Usage

```

createTable1CovariateSettings(
  specifications = getDefaultTable1Specifications(),
  covariateSettings = createDefaultCovariateSettings(),

```

```

    includedCovariateConceptIds = c(),
    addDescendantsToInclude = FALSE,
    excludedCovariateConceptIds = c(),
    addDescendantsToExclude = FALSE,
    includedCovariateIds = c()
  )

```

Arguments

specifications A specifications object for generating a table using the [createTable1](#) function.

covariateSettings

The covariate settings object to use as the basis for the filtered covariate settings.

includedCovariateConceptIds

A list of concept IDs that should be used to construct covariates.

addDescendantsToInclude

Should descendant concept IDs be added to the list of concepts to include?

excludedCovariateConceptIds

A list of concept IDs that should NOT be used to construct covariates.

addDescendantsToExclude

Should descendant concept IDs be added to the list of concepts to exclude?

includedCovariateIds

A list of covariate IDs that should be restricted to.

Value

A covariate settings object, for example to be used when calling the [getDbCovariateData](#) function.

Examples

```

table1CovSettings <- createTable1CovariateSettings(
  specifications = getDefaultTable1Specifications(),
  covariateSettings = createDefaultCovariateSettings(),
  includedCovariateConceptIds = c(),
  addDescendantsToInclude = FALSE,
  excludedCovariateConceptIds = c(),
  addDescendantsToExclude = FALSE,
  includedCovariateIds = c()
)

```

createTemporalCovariateSettings

Create covariate settings

Description

Create covariate settings

Usage

```

createTemporalCovariateSettings(
  useDemographicsGender = FALSE,
  useDemographicsAge = FALSE,
  useDemographicsAgeGroup = FALSE,
  useDemographicsRace = FALSE,
  useDemographicsEthnicity = FALSE,
  useDemographicsIndexYear = FALSE,
  useDemographicsIndexMonth = FALSE,
  useDemographicsPriorObservationTime = FALSE,
  useDemographicsPostObservationTime = FALSE,
  useDemographicsTimeInCohort = FALSE,
  useDemographicsIndexYearMonth = FALSE,
  useCareSiteId = FALSE,
  useConditionOccurrence = FALSE,
  useConditionOccurrencePrimaryInpatient = FALSE,
  useConditionEraStart = FALSE,
  useConditionEraOverlap = FALSE,
  useConditionEraGroupStart = FALSE,
  useConditionEraGroupOverlap = FALSE,
  useDrugExposure = FALSE,
  useDrugEraStart = FALSE,
  useDrugEraOverlap = FALSE,
  useDrugEraGroupStart = FALSE,
  useDrugEraGroupOverlap = FALSE,
  useProcedureOccurrence = FALSE,
  useDeviceExposure = FALSE,
  useMeasurement = FALSE,
  useMeasurementValue = FALSE,
  useMeasurementRangeGroup = FALSE,
  useMeasurementValueAsConcept = FALSE,
  useObservation = FALSE,
  useObservationValueAsConcept = FALSE,
  useObservationValue = FALSE,
  useCharlsonIndex = FALSE,
  useDcsi = FALSE,
  useChads2 = FALSE,
  useChads2Vasc = FALSE,
  useHfrs = FALSE,
  useDistinctConditionCount = FALSE,
  useDistinctIngredientCount = FALSE,
  useDistinctProcedureCount = FALSE,
  useDistinctMeasurementCount = FALSE,
  useDistinctObservationCount = FALSE,
  useVisitCount = FALSE,
  useVisitConceptCount = FALSE,
  temporalStartDays = -365:-1,
  temporalEndDays = -365:-1,
  includedCovariateConceptIds = c(),
  addDescendantsToInclude = FALSE,
  excludedCovariateConceptIds = c(),
  addDescendantsToExclude = FALSE,

```

```

    includedCovariateIds = c()
  )

```

Arguments

`useDemographicsGender`
Gender of the subject. (analysis ID 1)

`useDemographicsAge`
Age of the subject on the index date (in years). (analysis ID 2)

`useDemographicsAgeGroup`
Age of the subject on the index date (in 5 year age groups) (analysis ID 3)

`useDemographicsRace`
Race of the subject. (analysis ID 4)

`useDemographicsEthnicity`
Ethnicity of the subject. (analysis ID 5)

`useDemographicsIndexYear`
Year of the index date. (analysis ID 6)

`useDemographicsIndexMonth`
Month of the index date. (analysis ID 7)

`useDemographicsPriorObservationTime`
Number of days of observation time preceding the index date. (analysis ID 8)

`useDemographicsPostObservationTime`
Number of days of observation time preceding the index date. (analysis ID 9)

`useDemographicsTimeInCohort`
Number of days of observation time preceding the index date. (analysis ID 10)

`useDemographicsIndexYearMonth`
Calendar month of the index date. (analysis ID 11)

`useCareSiteId` Care site associated with the cohort start, pulled from the `visit_detail`, `visit_occurrence`, or `person` table, in that order. (analysis ID 12)

`useConditionOccurrence`
One covariate per condition in the `condition_occurrence` table starting in the time window. (analysis ID 101)

`useConditionOccurrencePrimaryInpatient`
One covariate per condition observed as a primary diagnosis in an inpatient setting in the `condition_occurrence` table starting in the time window. (analysis ID 102)

`useConditionEraStart`
One covariate per condition in the `condition_era` table starting in the time window. (analysis ID 201)

`useConditionEraOverlap`
One covariate per condition in the `condition_era` table overlapping with any part of the time window. (analysis ID 202)

`useConditionEraGroupStart`
One covariate per condition era rolled up to SNOMED groups in the `condition_era` table starting in the time window. (analysis ID 203)

`useConditionEraGroupOverlap`
One covariate per condition era rolled up to SNOMED groups in the `condition_era` table overlapping with any part of the time window. (analysis ID 204)

useDrugExposure	One covariate per drug in the drug_exposure table starting in the time window. (analysis ID 301)
useDrugEraStart	One covariate per drug in the drug_era table starting in the time window. (analysis ID 401)
useDrugEraOverlap	One covariate per drug in the drug_era table overlapping with any part of the time window. (analysis ID 402)
useDrugEraGroupStart	One covariate per drug rolled up to ATC groups in the drug_era table starting in the time window. (analysis ID 403)
useDrugEraGroupOverlap	One covariate per drug rolled up to ATC groups in the drug_era table overlapping with any part of the time window. (analysis ID 404)
useProcedureOccurrence	One covariate per procedure in the procedure_occurrence table in the time window. (analysis ID 501)
useDeviceExposure	One covariate per device in the device exposure table starting in the timewindow. (analysis ID 601)
useMeasurement	One covariate per measurement in the measurement table in the time window. (analysis ID 701)
useMeasurementValue	One covariate containing the value per measurement-unit combination in the time window. If multiple values are found, the last is taken. (analysis ID 702)
useMeasurementRangeGroup	Covariates indicating whether measurements are below, within, or above normal range within the time period. (analysis ID 703)
useMeasurementValueAsConcept	One covariate per measurement-value concept combination within the time period. (analysis ID 704)
useObservation	One covariate per observation in the observation table in the time window. (analysis ID 801)
useObservationValueAsConcept	One covariate per observation-value concept combination within the time period. (analysis ID 802)
useObservationValue	One covariate containing the value per observation-unit combination in the time window. If multiple values are found, the last is taken. (analysis ID 803)
useCharlsonIndex	The Charlson comorbidity index (Romano adaptation) using all conditions prior to the window end. (analysis ID 901)
useDcsi	The Diabetes Comorbidity Severity Index (DCSI) using all conditions prior to the window end. (analysis ID 902)
useChads2	The CHADS2 score using all conditions prior to the window end. (analysis ID 903)
useChads2Vasc	The CHADS2VAsC score using all conditions prior to the window end. (analysis ID 904)

<code>useHfrs</code>	The Hospital Frailty Risk Score score using all conditions prior to the window end. (analysis ID 926)
<code>useDistinctConditionCount</code>	The number of distinct condition concepts observed in the time window. (analysis ID 905)
<code>useDistinctIngredientCount</code>	The number of distinct ingredients observed in the time window. (analysis ID 906)
<code>useDistinctProcedureCount</code>	The number of distinct procedures observed in the time window. (analysis ID 907)
<code>useDistinctMeasurementCount</code>	The number of distinct measurements observed in the time window. (analysis ID 908)
<code>useDistinctObservationCount</code>	The number of distinct observations in the time window. (analysis ID 909)
<code>useVisitCount</code>	The number of visits observed in the time window. (analysis ID 910)
<code>useVisitConceptCount</code>	The number of visits observed in the time window, stratified by visit concept ID. (analysis ID 911)
<code>temporalStartDays</code>	A list of integers representing the start of a time period, relative to the index date. 0 indicates the index date, -1 indicates the day before the index date, etc. The start day is included in the time period.
<code>temporalEndDays</code>	A list of integers representing the end of a time period, relative to the index date. 0 indicates the index date, -1 indicates the day before the index date, etc. The end day is included in the time period.
<code>includedCovariateConceptIds</code>	A list of concept IDs that should be used to construct covariates.
<code>addDescendantsToInclude</code>	Should descendant concept IDs be added to the list of concepts to include?
<code>excludedCovariateConceptIds</code>	A list of concept IDs that should NOT be used to construct covariates.
<code>addDescendantsToExclude</code>	Should descendant concept IDs be added to the list of concepts to exclude?
<code>includedCovariateIds</code>	A list of covariate IDs that should be restricted to.

Details

creates an object specifying how covariates should be constructed from data in the CDM model.

Value

An object of type `covariateSettings`, to be used in other functions.

Examples

```

settings <- createTemporalCovariateSettings(
  useDemographicsGender = TRUE,
  useDemographicsAge = FALSE,
  useDemographicsAgeGroup = TRUE,
  useDemographicsRace = TRUE,
  useDemographicsEthnicity = TRUE,
  useDemographicsIndexYear = TRUE,
  useDemographicsIndexMonth = TRUE,
  useDemographicsPriorObservationTime = FALSE,
  useDemographicsPostObservationTime = FALSE,
  useDemographicsTimeInCohort = FALSE,
  useDemographicsIndexYearMonth = FALSE,
  useCareSiteId = FALSE,
  useConditionOccurrence = FALSE,
  useConditionOccurrencePrimaryInpatient = FALSE,
  useConditionEraStart = FALSE,
  useConditionEraOverlap = FALSE,
  useConditionEraGroupStart = FALSE,
  useConditionEraGroupOverlap = TRUE,
  useDrugExposure = FALSE,
  useDrugEraStart = FALSE,
  useDrugEraOverlap = FALSE,
  useDrugEraGroupStart = FALSE,
  useDrugEraGroupOverlap = TRUE,
  useProcedureOccurrence = TRUE,
  useDeviceExposure = TRUE,
  useMeasurement = TRUE,
  useMeasurementValue = FALSE,
  useMeasurementRangeGroup = TRUE,
  useMeasurementValueAsConcept = TRUE,
  useObservation = TRUE,
  useObservationValueAsConcept = TRUE,
  useObservationValue = FALSE,
  useCharlsonIndex = TRUE,
  useDcsi = TRUE,
  useChads2 = TRUE,
  useChads2Vasc = TRUE,
  useHfrs = FALSE,
  useDistinctConditionCount = FALSE,
  useDistinctIngredientCount = FALSE,
  useDistinctProcedureCount = FALSE,
  useDistinctMeasurementCount = FALSE,
  useDistinctObservationCount = FALSE,
  useVisitCount = FALSE,
  useVisitConceptCount = FALSE,
  temporalStartDays = -365:-1,
  temporalEndDays = -365:-1,
  includedCovariateConceptIds = c(),
  addDescendantsToInclude = FALSE,
  excludedCovariateConceptIds = c(),
  addDescendantsToExclude = FALSE,
  includedCovariateIds = c()
)

```

```
createTemporalSequenceCovariateSettings
    Create covariate settings
```

Description

Create covariate settings

Usage

```
createTemporalSequenceCovariateSettings(
  useDemographicsGender = FALSE,
  useDemographicsAge = FALSE,
  useDemographicsAgeGroup = FALSE,
  useDemographicsRace = FALSE,
  useDemographicsEthnicity = FALSE,
  useDemographicsIndexYear = FALSE,
  useDemographicsIndexMonth = FALSE,
  useConditionOccurrence = FALSE,
  useConditionOccurrencePrimaryInpatient = FALSE,
  useConditionEraStart = FALSE,
  useConditionEraGroupStart = FALSE,
  useDrugExposure = FALSE,
  useDrugEraStart = FALSE,
  useDrugEraGroupStart = FALSE,
  useProcedureOccurrence = FALSE,
  useDeviceExposure = FALSE,
  useMeasurement = FALSE,
  useMeasurementValue = FALSE,
  useObservation = FALSE,
  timePart = "month",
  timeInterval = 1,
  sequenceEndDay = -1,
  sequenceStartDay = -730,
  includedCovariateConceptIds = c(),
  addDescendantsToInclude = FALSE,
  excludedCovariateConceptIds = c(),
  addDescendantsToExclude = FALSE,
  includedCovariateIds = c()
)
```

Arguments

```
useDemographicsGender
    Gender of the subject. (analysis ID 1)
useDemographicsAge
    Age of the subject on the index date (in years). (analysis ID 2)
useDemographicsAgeGroup
    Age of the subject on the index date (in 5 year age groups) (analysis ID 3)
useDemographicsRace
    Race of the subject. (analysis ID 4)
```

useDemographicsEthnicity	Ethnicity of the subject. (analysis ID 5)
useDemographicsIndexYear	Year of the index date. (analysis ID 6)
useDemographicsIndexMonth	Month of the index date. (analysis ID 7)
useConditionOccurrence	One covariate per condition in the condition_occurrence table starting in the time window. (analysis ID 101)
useConditionOccurrencePrimaryInpatient	One covariate per condition observed as a primary diagnosis in an inpatient setting in the condition_occurrence table starting in the time window. (analysis ID 102)
useConditionEraStart	One covariate per condition in the condition_era table starting in the time window. (analysis ID 201)
useConditionEraGroupStart	One covariate per condition era rolled up to SNOMED groups in the condition_era table starting in the time window. (analysis ID 203)
useDrugExposure	One covariate per drug in the drug_exposure table starting in the time window. (analysis ID 301)
useDrugEraStart	One covariate per drug in the drug_era table starting in the time window. (analysis ID 401)
useDrugEraGroupStart	One covariate per drug rolled up to ATC groups in the drug_era table starting in the time window. (analysis ID 403)
useProcedureOccurrence	One covariate per procedure in the procedure_occurrence table in the time window. (analysis ID 501)
useDeviceExposure	One covariate per device in the device exposure table starting in the timewindow. (analysis ID 601)
useMeasurement	One covariate per measurement in the measurement table in the time window. (analysis ID 701)
useMeasurementValue	One covariate containing the value per measurement-unit combination in the time window. If multiple values are found, the last is taken. (analysis ID 702)
useObservation	One covariate per observation in the observation table in the time window. (analysis ID 801)
timePart	The interval scale ('DAY', 'MONTH', 'YEAR')
timeInterval	Fixed interval length for timeId using the 'timePart' scale. For example, a 'timePart' of DAY with 'timeInterval' 30 has timeIds where timeId 1 is day 0 to day 29, timeId 2 is day 30 to day 59, etc.
sequenceEndDay	What is the end day (relative to the index date) of the data extraction?
sequenceStartDay	What is the start day (relative to the index date) of the data extraction?

`includedCovariateConceptIds`
 A list of concept IDs that should be used to construct covariates.

`addDescendantsToInclude`
 Should descendant concept IDs be added to the list of concepts to include?

`excludedCovariateConceptIds`
 A list of concept IDs that should NOT be used to construct covariates.

`addDescendantsToExclude`
 Should descendant concept IDs be added to the list of concepts to exclude?

`includedCovariateIds`
 A list of covariate IDs that should be restricted to.

Details

creates an object specifying how covariates should be constructed from data in the CDM model.

Value

An object of type `covariateSettings`, to be used in other functions.

Examples

```
settings <- createTemporalSequenceCovariateSettings(  
  useDemographicsGender = TRUE,  
  useDemographicsAge = FALSE,  
  useDemographicsAgeGroup = TRUE,  
  useDemographicsRace = TRUE,  
  useDemographicsEthnicity = TRUE,  
  useDemographicsIndexYear = TRUE,  
  useDemographicsIndexMonth = TRUE,  
  useConditionOccurrence = FALSE,  
  useConditionOccurrencePrimaryInpatient = FALSE,  
  useConditionEraStart = FALSE,  
  useConditionEraGroupStart = FALSE,  
  useDrugExposure = FALSE,  
  useDrugEraStart = FALSE,  
  useDrugEraGroupStart = FALSE,  
  useProcedureOccurrence = TRUE,  
  useDeviceExposure = TRUE,  
  useMeasurement = TRUE,  
  useMeasurementValue = FALSE,  
  useObservation = TRUE,  
  timePart = "DAY",  
  timeInterval = 1,  
  sequenceEndDay = -1,  
  sequenceStartDay = -730,  
  includedCovariateConceptIds = c(),  
  addDescendantsToInclude = FALSE,  
  excludedCovariateConceptIds = c(),  
  addDescendantsToExclude = FALSE,  
  includedCovariateIds = c()  
)
```

`filterByCohortDefinitionId`*Filter covariates by cohort definition IDs*

Description

Filter covariates by cohort definition IDs

Usage

```
filterByCohortDefinitionId(covariateData, cohortId = 1, cohortIds = c(1))
```

Arguments

<code>covariateData</code>	An object of type <code>CovariateData</code>
<code>cohortId</code>	DEPRECATED The cohort definition IDs to keep.
<code>cohortIds</code>	The cohort definition IDs to keep.

Value

An object of type `covariateData`.

Examples

```
covariateData <- FeatureExtraction::createEmptyCovariateData(  
  cohortIds = c(1, 2),  
  aggregated = TRUE,  
  temporal = FALSE  
)  
  
covData <- filterByCohortDefinitionId(  
  covariateData = covariateData,  
  cohortIds = c(1)  
)
```

`filterByRowId`*Filter covariates by row ID*

Description

Filter covariates by row ID

Usage

```
filterByRowId(covariateData, rowIds)
```

Arguments

covariateData An object of type CovariateData
 rowIds A vector containing the rowIds to keep.

Value

An object of type covariateData.

Examples

```
covariateData <- FeatureExtraction::createEmptyCovariateData(
  cohortIds = 1,
  aggregated = FALSE,
  temporal = FALSE
)

covData <- filterByRowId(
  covariateData = covariateData,
  rowIds = 1
)
```

getDbCohortAttrCovariatesData

Getcovariate information from the database through the cohort_attribute table

Description

Constructs covariates using the cohort_attribute table.

Usage

```
getDbCohortAttrCovariatesData(
  connection,
  oracleTempSchema = NULL,
  cdmDatabaseSchema,
  cohortTable = "#cohort_person",
  cohortId = -1,
  cohortIds = c(-1),
  cdmVersion = "5",
  rowIdField = "subject_id",
  covariateSettings,
  targetDatabaseSchema = NULL,
  targetCovariateTable = NULL,
  targetCovariateContinuousTable = NULL,
  targetCovariateRefTable = NULL,
  targetAnalysisRefTable = NULL,
  targetTimeRefTable = NULL,
  aggregated = FALSE,
```

```

    minCharacterizationMean = 0,
    minCharacterizationCount = 0,
    tempEmulationSchema = getOption("sqlRenderTempEmulationSchema")
)

```

Arguments

connection	A connection to the server containing the schema as created using the connect function in the DatabaseConnector package.
oracleTempSchema	DEPRECATED: use tempEmulationSchema instead.
cdmDatabaseSchema	The name of the database schema that contains the OMOP CDM instance. Requires read permissions to this database. On SQL Server, this should specify both the database and the schema, so for example 'cdm_instance.dbo'.
cohortTable	Name of the table holding the cohort for which we want to construct covariates. If it is a temp table, the name should have a hash prefix, e.g. '#temp_table'. If it is a non-temp table, it should include the database schema, e.g. 'cdm_database.cohort'.
cohortId	DEPRECATED: For which cohort ID should covariates be constructed? If set to -1, covariates will be constructed for all cohorts in the specified cohort table.
cohortIds	For which cohort ID(s) should covariates be constructed? If set to c(-1), covariates will be constructed for all cohorts in the specified cohort table.
cdmVersion	The version of the Common Data Model used. Currently only cdmVersion = "5" is supported.
rowIdField	The name of the field in the cohort temp table that is to be used as the row_id field in the output table. This can be especially useful if there is more than one period per person.
covariateSettings	An object of type covariateSettings as created using the createCohortAttrCovariateSettings function.
targetDatabaseSchema	Optional target database schema. Ignored because cohort attribute covariates do not support exporting directly to target tables.
targetCovariateTable	Optional target covariate table. Ignored because cohort attribute covariates do not support exporting directly to target tables.
targetCovariateContinuousTable	Optional target continuous covariate table. Ignored because cohort attribute covariates do not support exporting directly to target tables.
targetCovariateRefTable	Optional target covariate reference table. Ignored because cohort attribute covariates do not support exporting directly to target tables.
targetAnalysisRefTable	Optional target analysis reference table. Ignored because cohort attribute covariates do not support exporting directly to target tables.
targetTimeRefTable	Optional target time reference table. Ignored because cohort attribute covariates do not support exporting directly to target tables.
aggregated	Should aggregate statistics be computed instead of covariates per cohort entry?

minCharacterizationMean

The minimum mean value for characterization output. Ignored because aggregation is not supported for cohort attribute covariates.

minCharacterizationCount

The minimum count value for characterization output. Ignored because aggregation is not supported for cohort attribute covariates.

tempEmulationSchema

Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.

Details

This function uses the data in the CDM to construct a large set of covariates for the provided cohort. The cohort is assumed to be in an existing temp table with these fields: 'subject_id', 'cohort_definition_id', 'cohort_start_date'. Optionally, an extra field can be added containing the unique identifier that will be used as rowID in the output. Typically, users don't call this function directly but rather use the [getDbCovariateData](#) function instead.

Value

Returns an object of type `CovariateData`, which is an Andromeda object containing information on the baseline covariates. Information about multiple outcomes can be captured at once for efficiency reasons. This object is a list with the following components:

covariates An `ffdf` object listing the baseline covariates per person in the cohorts. This is done using a sparse representation: covariates with a value of 0 are omitted to save space. The covariates object will have three columns: `rowId`, `covariateId`, and `covariateValue`. The `rowId` is usually equal to the `person_id`, unless specified otherwise in the `rowIdField` argument.

covariateRef A table describing the covariates that have been extracted.

. The `CovariateData` object will also have a `metaData` attribute, a list of objects with information on how the covariateData object was constructed.

Examples

```
connectionDetails <- Eunomia::getEunomiaConnectionDetails()
Eunomia::createCohorts(
  connectionDetails = connectionDetails,
  cdmDatabaseSchema = "main",
  cohortDatabaseSchema = "main",
  cohortTable = "cohort"
)
connection <- DatabaseConnector::connect(connectionDetails)
covariateSettings <- createCohortAttrCovariateSettings(
  attrDatabaseSchema = "main",
  cohortAttrTable = "cohort_attribute",
  attrDefinitionTable = "attribute_definition",
  includeAttrIds = c(1),
  isBinary = FALSE,
  missingMeansZero = FALSE
)

covData <- getDbCohortAttrCovariatesData(
  connection = connection,
```

```

tempEmulationSchema = NULL,
cdmDatabaseSchema = "main",
cdmVersion = "5",
cohortTable = "cohort",
cohortIds = 1,
rowIdField = "subject_id",
covariateSettings = covariateSettings,
aggregated = FALSE
)

```

getDbCohortBasedCovariatesData

Get covariate information from the database based on other cohorts

Description

Constructs covariates using other cohorts.

Usage

```

getDbCohortBasedCovariatesData(
  connection,
  oracleTempSchema = NULL,
  cdmDatabaseSchema,
  cohortTable = "#cohort_person",
  cohortId = -1,
  cohortIds = c(-1),
  cdmVersion = "5",
  rowIdField = "subject_id",
  covariateSettings,
  targetDatabaseSchema = NULL,
  targetCovariateTable = NULL,
  targetCovariateContinuousTable = NULL,
  targetCovariateRefTable = NULL,
  targetAnalysisRefTable = NULL,
  targetTimeRefTable = NULL,
  aggregated = FALSE,
  minCharacterizationMean = 0,
  minCharacterizationCount = 0,
  tempEmulationSchema = getOption("sqlRenderTempEmulationSchema")
)

```

Arguments

connection	A connection to the server containing the schema as created using the connect function in the DatabaseConnector package.
oracleTempSchema	DEPRECATED: use tempEmulationSchema instead.

cdmDatabaseSchema	The name of the database schema that contains the OMOP CDM instance. Requires read permissions to this database. On SQL Server, this should specify both the database and the schema, so for example 'cdm_instance.dbo'.
cohortTable	Name of the table holding the cohort for which we want to construct covariates. If it is a temp table, the name should have a hash prefix, e.g. '#temp_table'. If it is a non-temp table, it should include the database schema, e.g. 'cdm_database.cohort'.
cohortId	DEPRECATED:For which cohort ID should covariates be constructed? If set to -1, covariates will be constructed for all cohorts in the specified cohort table.
cohortIds	For which cohort ID(s) should covariates be constructed? If set to c(-1), covariates will be constructed for all cohorts in the specified cohort table.
cdmVersion	The version of the Common Data Model used. Currently only cdmVersion = "5" is supported.
rowIdField	The name of the field in the cohort temp table that is to be used as the row_id field in the output table. This can be especially usefull if there is more than one period per person.
covariateSettings	An object of type covariateSettings as created using the createCohortBasedCovariateSetting or createCohortBasedTemporalCovariateSettings functions.
targetDatabaseSchema	(Optional) The name of the database schema where the resulting covariates should be stored. If not provided, results will be fetched to R.
targetCovariateTable	(Optional) The name of the table where the resulting covariates will be stored. If not provided, results will be fetched to R. The table can be a permanent table in the targetDatabaseSchema or a temp table. If it is a temp table, do not specify targetDatabaseSchema.
targetCovariateContinuousTable	(Optional) The name of the table where the resulting continuous covariates should be stored.
targetCovariateRefTable	(Optional) The name of the table where the covariate reference will be stored.
targetAnalysisRefTable	(Optional) The name of the table where the analysis reference will be stored.
targetTimeRefTable	(Optional) The name of the table for the time reference
aggregated	Should aggregate statistics be computed instead of covariates per cohort entry?
minCharacterizationMean	The minimum mean value for binary characterization output. Values below this will be cut off from output. This will help reduce the file size of the characterization output, but will remove information on covariates that have very low values. The default is 0.
minCharacterizationCount	The minimum count value for binary characterization output. Values below this will be cut off from output. This will help reduce the file size of the characterization output, but will remove information on covariates that occur in very few cohort entries. The default is 0.
tempEmulationSchema	Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.

Details

This function uses the data in the CDM to construct a large set of covariates for the provided cohort. The cohort is assumed to be in an existing temp table with these fields: 'subject_id', 'cohort_definition_id', 'cohort_start_date'. Optionally, an extra field can be added containing the unique identifier that will be used as rowID in the output. Typically, users don't call this function directly but rather use the [getDbCovariateData](#) function instead.

Value

Returns an object of type `CovariateData`, which is an Andromeda object containing information on the baseline covariates. Information about multiple outcomes can be captured at once for efficiency reasons. This object is a list with the following components:

covariates An `ffdf` object listing the baseline covariates per person in the cohorts. This is done using a sparse representation: covariates with a value of 0 are omitted to save space. The covariates object will have three columns: `rowId`, `covariateId`, and `covariateValue`. The `rowId` is usually equal to the `person_id`, unless specified otherwise in the `rowIdField` argument.

covariateRef A table describing the covariates that have been extracted.

. The `CovariateData` object will also have a `metaData` attribute, a list of objects with information on how the covariateData object was constructed.

<code>getDbCovariateData</code>	<i>Get covariate information from the database</i>
---------------------------------	--

Description

Uses one or several covariate builder functions to construct covariates.

Usage

```
getDbCovariateData(
  connectionDetails = NULL,
  connection = NULL,
  oracleTempSchema = NULL,
  cdmDatabaseSchema,
  cdmVersion = "5",
  cohortTable = "cohort",
  cohortDatabaseSchema = cdmDatabaseSchema,
  cohortTableIsTemp = FALSE,
  cohortId = -1,
  cohortIds = c(-1),
  rowIdField = "subject_id",
  covariateSettings,
  exportToTable = FALSE,
  createTable = exportToTable,
  dropTableIfExists = exportToTable,
  targetDatabaseSchema = NULL,
  targetCovariateTable = NULL,
  targetCovariateContinuousTable = NULL,
  targetCovariateRefTable = NULL,
```

```

targetAnalysisRefTable = NULL,
targetTimeRefTable = NULL,
aggregated = FALSE,
minCharacterizationMean = 0,
minCharacterizationCount = 0,
tempEmulationSchema = getOption("sqlRenderTempEmulationSchema"),
covariateCohortDatabaseSchema = NULL,
covariateCohortTable = NULL
)

```

Arguments

connectionDetails

An R object of type `connectionDetails` created using the function `createConnectionDetails` in the `DatabaseConnector` package. Either the `connection` or `connectionDetails` argument should be specified.

connection

A connection to the server containing the schema as created using the `connect` function in the `DatabaseConnector` package. Either the `connection` or `connectionDetails` argument should be specified.

oracleTempSchema

DEPRECATED: use `tempEmulationSchema` instead.

cdmDatabaseSchema

The name of the database schema that contains the OMOP CDM instance. Requires read permissions to this database. On SQL Server, this should specify both the database and the schema, so for example `'cdm_instance.dbo'`.

cdmVersion

Define the OMOP CDM version used: currently supported is "5".

cohortTable

Name of the (temp) table holding the cohort for which we want to construct covariates

cohortDatabaseSchema

If the cohort table is not a temp table, specify the database schema where the cohort table can be found. On SQL Server, this should specify both the database and the schema, so for example `'cdm_instance.dbo'`.

cohortTableIsTemp

Is the cohort table a temp table?

cohortId

DEPRECATED: For which cohort ID(s) should covariates be constructed? If set to -1, covariates will be constructed for all cohorts in the specified cohort table.

cohortIds

For which cohort ID(s) should covariates be constructed? If set to `c(-1)`, covariates will be constructed for all cohorts in the specified cohort table.

rowIdField

The name of the field in the cohort table that is to be used as the `row_id` field in the output table. This can be especially useful if there is more than one period per person.

covariateSettings

Either an object of type `covariateSettings` as created using one of the `createCovariate` functions, or a list of such objects.

exportToTable

Whether to export to a table rather than Andromeda object

createTable

Run sql to create table? Code does not check if table exists.

dropTableIfExists

If `targetDatabaseSchema`, drop any existing tables. Otherwise, results are merged into existing table data. Overrides `createTable`.

targetDatabaseSchema	(Optional) The name of the database schema where the resulting covariates should be stored as a table. If not provided, results will be fetched to R.
targetCovariateTable	(Optional) The name of the table where the resulting covariates will be stored. If not provided, results will be fetched to R. The table can be a permanent table in the targetDatabaseSchema or a temp table. If it is a temp table, do not specify targetDatabaseSchema.
targetCovariateContinuousTable	(Optional) The name of the table where the resulting continuous covariates will be stored. If not provided, results will be fetched to R. The table can be a permanent table in the targetDatabaseSchema or a temp table. If it is a temp table, do not specify targetDatabaseSchema.
targetCovariateRefTable	(Optional) The name of the table where the covariate reference will be stored. If it is a temp table, do not specify targetDatabaseSchema.
targetAnalysisRefTable	(Optional) The name of the table where the analysis reference will be stored. If it is a temp table, do not specify targetDatabaseSchema.
targetTimeRefTable	(Optional) The name of the table for the time reference. If it is a temp table, do not specify targetDatabaseSchema.
aggregated	Should aggregate statistics be computed instead of covariates per cohort entry? If aggregated is set to FALSE, the results returned will be based on each subject_id and cohort_start_date in your cohort table. If your cohort contains multiple entries for the same subject_id (due to different cohort_start_date values), you must carefully set the rowIdField so you can identify the patients properly. See issue #229 for more discussion on this parameter.
minCharacterizationMean	The minimum mean value for characterization output. Values below this will be cut off from output. This will help reduce the file size of the characterization output, but will remove information on covariates that have very low values. The default is 0.
minCharacterizationCount	The minimum count value for characterization output. Values below this will be cut off from output. This will help reduce the file size of the characterization output, but will remove information on covariates that occur in very few cohort entries. The default is 0.
tempEmulationSchema	Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.
covariateCohortDatabaseSchema	The database schema where the cohorts used to define the covariates can be found.
covariateCohortTable	The table where the cohorts used to define the covariates can be found.

Details

This function uses the data in the CDM to construct a large set of covariates for the provided cohort. The cohort is assumed to be in an existing table with these fields: 'subject_id', 'co-

hort_definition_id', 'cohort_start_date'. Optionally, an extra field can be added containing the unique identifier that will be used as rowID in the output.

Value

Returns an object of type covariateData, containing information on the covariates.

Examples

```
eunomiaConnectionDetails <- Eunomia::getEunomiaConnectionDetails()
covSettings <- createDefaultCovariateSettings()
Eunomia::createCohorts(
  connectionDetails = eunomiaConnectionDetails,
  cdmDatabaseSchema = "main",
  cohortDatabaseSchema = "main",
  cohortTable = "cohort"
)
covData <- getDbCovariateData(
  connectionDetails = eunomiaConnectionDetails,
  tempEmulationSchema = NULL,
  cdmDatabaseSchema = "main",
  cdmVersion = "5",
  cohortTable = "cohort",
  cohortDatabaseSchema = "main",
  cohortTableIsTemp = FALSE,
  cohortIds = -1,
  rowIdField = "subject_id",
  covariateSettings = covSettings,
  aggregated = FALSE
)
```

getDbDefaultCovariateData

Get default covariate information from the database

Description

Constructs a large default set of covariates for one or more cohorts using data in the CDM schema. Includes covariates for all drugs, drug classes, condition, condition classes, procedures, observations, etc.

Usage

```
getDbDefaultCovariateData(
  connection,
  oracleTempSchema = NULL,
  cdmDatabaseSchema,
  cohortTable = "#cohort_person",
  cohortId = -1,
  cohortIds = c(-1),
  cdmVersion = "5",
```

```

    rowIdField = "subject_id",
    covariateSettings,
    targetDatabaseSchema = NULL,
    targetCovariateTable = NULL,
    targetCovariateContinuousTable = NULL,
    targetCovariateRefTable = NULL,
    targetAnalysisRefTable = NULL,
    targetTimeRefTable = NULL,
    aggregated = FALSE,
    minCharacterizationMean = 0,
    minCharacterizationCount = 0,
    tempEmulationSchema = getOption("sqlRenderTempEmulationSchema")
)

```

Arguments

connection	A connection to the server containing the schema as created using the connect function in the DatabaseConnector package.
oracleTempSchema	DEPRECATED: use tempEmulationSchema instead.
cdmDatabaseSchema	The name of the database schema that contains the OMOP CDM instance. Requires read permissions to this database. On SQL Server, this should specify both the database and the schema, so for example 'cdm_instance.dbo'.
cohortTable	Name of the table holding the cohort for which we want to construct covariates. If it is a temp table, the name should have a hash prefix, e.g. '#temp_table'. If it is a non-temp table, it should include the database schema, e.g. 'cdm_database.cohort'.
cohortId	DEPRECATED: For which cohort ID should covariates be constructed? If set to -1, covariates will be constructed for all cohorts in the specified cohort table.
cohortIds	For which cohort ID(s) should covariates be constructed? If set to c(-1), covariates will be constructed for all cohorts in the specified cohort table.
cdmVersion	The version of the Common Data Model used. Currently only cdmVersion = "5" is supported.
rowIdField	The name of the field in the cohort temp table that is to be used as the row_id field in the output table. This can be especially useful if there is more than one period per person.
covariateSettings	Either an object of type covariateSettings as created using one of the createCovariate functions, or a list of such objects.
targetDatabaseSchema	(Optional) The name of the database schema where the resulting covariates should be stored. If not provided, results will be fetched to R.
targetCovariateTable	(Optional) The name of the table where the resulting covariates will be stored. If not provided, results will be fetched to R. The table can be a permanent table in the targetDatabaseSchema or a temp table. If it is a temp table, do not specify targetDatabaseSchema.
targetCovariateContinuousTable	(Optional) The name of the table where the resulting continuous covariates should be stored.

targetCovariateRefTable	(Optional) The name of the table where the covariate reference will be stored.
targetAnalysisRefTable	(Optional) The name of the table where the analysis reference will be stored.
targetTimeRefTable	(Optional) The name of the table for the time reference
aggregated	Should aggregate statistics be computed instead of covariates per cohort entry?
minCharacterizationMean	The minimum mean value for binary characterization output. Values below this will be cut off from output. This will help reduce the file size of the characterization output, but will remove information on covariates that have very low values. The default is 0.
minCharacterizationCount	The minimum count value for binary characterization output. Values below this will be cut off from output. This will help reduce the file size of the characterization output, but will remove information on covariates that occur in very few cohort entries. The default is 0.
tempEmulationSchema	Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.

Details

This function uses the data in the CDM to construct a large set of covariates for the provided cohort. The cohort is assumed to be in an existing temp table with these fields: 'subject_id', 'cohort_definition_id', 'cohort_start_date'. Optionally, an extra field can be added containing the unique identifier that will be used as rowID in the output. Typically, users don't call this function directly but rather use the [getDbCovariateData](#) function instead.

Value

Returns an object of type `CovariateData`, which is an Andromeda object containing information on the baseline covariates. Information about multiple outcomes can be captured at once for efficiency reasons. This object is a list with the following components:

covariates An `ffdf` object listing the baseline covariates per person in the cohorts. This is done using a sparse representation: covariates with a value of 0 are omitted to save space. The covariates object will have three columns: `rowId`, `covariateId`, and `covariateValue`. The `rowId` is usually equal to the `person_id`, unless specified otherwise in the `rowIdField` argument.

covariateRef A table describing the covariates that have been extracted.

. The `CovariateData` object will also have a `metaData` attribute, a list of objects with information on how the covariateData object was constructed.

Examples

```
connectionDetails <- Eunomia::getEunomiaConnectionDetails()
Eunomia::createCohorts(
  connectionDetails = connectionDetails,
  cdmDatabaseSchema = "main",
  cohortDatabaseSchema = "main",
  cohortTable = "cohort"
```

```

)
connection <- DatabaseConnector::connect(connectionDetails)

results <- getDbDefaultCovariateData(
  connection = connection,
  cdmDatabaseSchema = "main",
  cohortTable = "cohort",
  covariateSettings = createDefaultCovariateSettings()
)

```

```
getDefaultTable1Specifications
```

Get the default table 1 specifications

Description

Loads the default specifications for a table 1, to be used with the `createTable1` function.

Usage

```
getDefaultTable1Specifications()
```

Value

A specifications objects.

Examples

```
defaultTable1Specs <- getDefaultTable1Specifications()
```

```
isAggregatedCovariateData
```

Check whether covariate data is aggregated

Description

Check whether covariate data is aggregated

Usage

```
isAggregatedCovariateData(x)
```

Arguments

`x` The covariate data object to check.

Value

A logical value.

Examples

```

covariateData <- FeatureExtraction::createEmptyCovariateData(
  cohortIds = 1,
  aggregated = FALSE,
  temporal = FALSE
)
isAggrCovData <- isAggregatedCovariateData(covariateData)

```

isCovariateData	<i>Check whether an object is a CovariateData object</i>
-----------------	--

Description

Check whether an object is a CovariateData object

Usage

```
isCovariateData(x)
```

Arguments

x	The object to check.
---	----------------------

Value

A logical value.

Examples

```

binaryCovDataFile <- system.file("testdata/binaryCovariateData.zip",
  package = "FeatureExtraction"
)
covData <- loadCovariateData(binaryCovDataFile)
isCovData <- isCovariateData(covData)

```

isTemporalCovariateData	<i>Check whether covariate data is temporal</i>
-------------------------	---

Description

Check whether covariate data is temporal

Usage

```
isTemporalCovariateData(x)
```

Arguments

x The covariate data object to check.

Value

A logical value.

Examples

```
covariateData <- FeatureExtraction::createEmptyCovariateData(  
  cohortIds = 1,  
  aggregated = FALSE,  
  temporal = FALSE  
)  
isTempCovData <- isTemporalCovariateData(covariateData)
```

loadCovariateData	<i>Load the covariate data from a folder</i>
-------------------	--

Description

loadCovariateData loads an object of type covariateData from a folder in the file system.

Usage

```
loadCovariateData(file, readOnly)
```

Arguments

file The name of the folder containing the data.
readOnly DEPRECATED: If true, the data is opened read only.

Details

The data will be written to a set of files in the folder specified by the user.

Value

An object of class CovariateData.

Examples

```
binaryCovDataFile <- system.file("testdata/binaryCovariateData.zip",  
  package = "FeatureExtraction"  
)  
covData <- loadCovariateData(binaryCovDataFile)
```

```
replaceCovariateSettingsCohortSchemaTable
```

Utility function to set the cohort table & schema on createCohortBasedCovariateSettings with information from the execution settings

Description

Utility function to set the cohort table & schema on createCohortBasedCovariateSettings with information from the execution settings

Usage

```
replaceCovariateSettingsCohortSchemaTable(
  covariateSettings,
  covariateCohortDatabaseSchema,
  covariateCohortTable
)
```

Arguments

covariateSettings

An object of type covariateSettings

covariateCohortDatabaseSchema

The database schema where the cohorts used to define the covariates can be found.

covariateCohortTable

The table where the cohorts used to define the covariates can be found.

Value

An object of type covariateSettings

```
saveCovariateData
```

Save the covariate data to folder

Description

saveCovariateData saves an object of type covariateData to folder.

Usage

```
saveCovariateData(covariateData, file)
```

Arguments

covariateData An object of type covariateData as generated using getDbCovariateData.

file The name of the folder where the data will be written. The folder should not yet exist.

Details

The data will be written to a set of files in the folder specified by the user.

Value

No return value, called for side effects.

Examples

```
covariateData <- FeatureExtraction::createEmptyCovariateData(
  cohortIds = 1,
  aggregated = FALSE,
  temporal = FALSE
)
# For this example we'll use a temporary file location:
fileName <- tempfile()
saveCovariateData(covariateData = covariateData, file = fileName)
# Cleaning up the file used in this example:
unlink(fileName)
```

tidyCovariateData	<i>Tidy covariate data</i>
-------------------	----------------------------

Description

Tidy covariate data

Usage

```
tidyCovariateData(
  covariateData,
  minFraction = 0.001,
  normalize = TRUE,
  removeRedundancy = TRUE
)
```

Arguments

covariateData	An object as generated using the getDbCovariateData function.
minFraction	Minimum fraction of the population that should have a non-zero value for a covariate for that covariate to be kept. Set to 0 to don't filter on frequency.
normalize	Normalize the covariates? (dividing by the max).
removeRedundancy	Should redundant covariates be removed?

Details

Normalize covariate values by dividing by the max and/or remove redundant covariates and/or remove infrequent covariates. For temporal covariates, redundancy is evaluated per time ID.

Value

An object of class `CovariateData`.

Examples

```
covariateData <- FeatureExtraction::createEmptyCovariateData(  
  cohortIds = 1,  
  aggregated = FALSE,  
  temporal = FALSE  
)  
  
covData <- tidyCovariateData(  
  covariateData = covariateData,  
  minFraction = 0.001,  
  normalize = TRUE,  
  removeRedundancy = TRUE  
)
```

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